Graphs of circles

1 Every point on the circle is exactly 5 units away from the point $(0, 0)$.

2

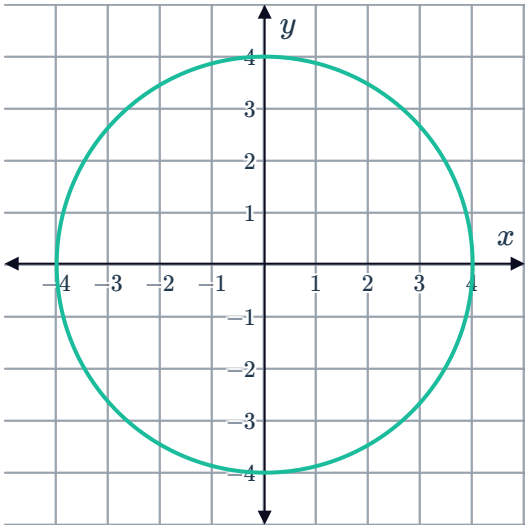
a Inside **b** On **c** Outside **d** On

3

a

i 4 units

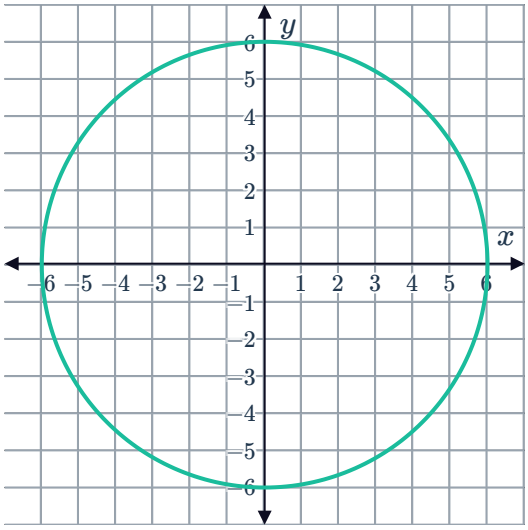
ii



b

i 6 units

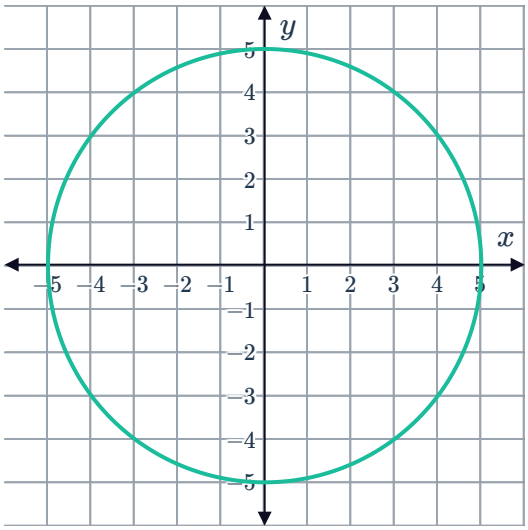
ii



c

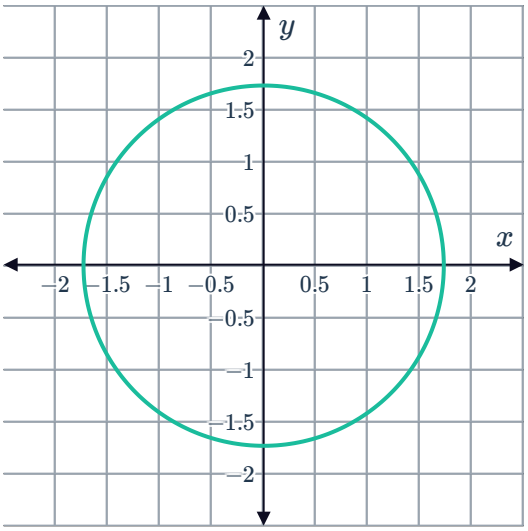
i 5 units

ii



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4



5

a

i $\left(-3.1, \frac{3}{2}\right)$ ii $\frac{3}{4}$ units

b

i $(6, 6)$ ii $2\sqrt{3}$ units

c

i $(0.7, -4.4)$ ii $\sqrt{3}$ units

d

i $(8, -1)$ ii $\sqrt{13}$ units

6

a

i $(0, 0)$ ii 2 units

b

i $(3, 2)$ ii 3 units

c

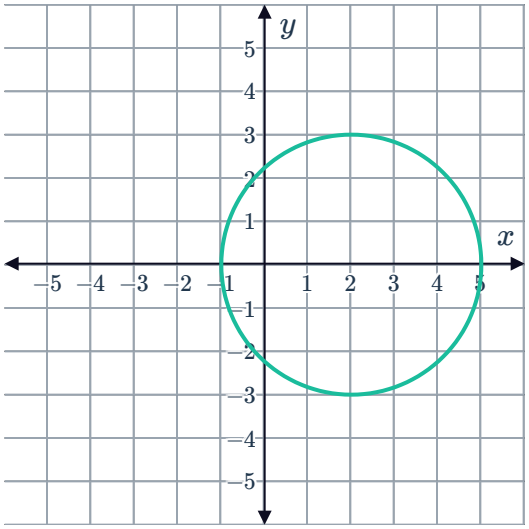
i $(1, -2)$ ii 8 units

d

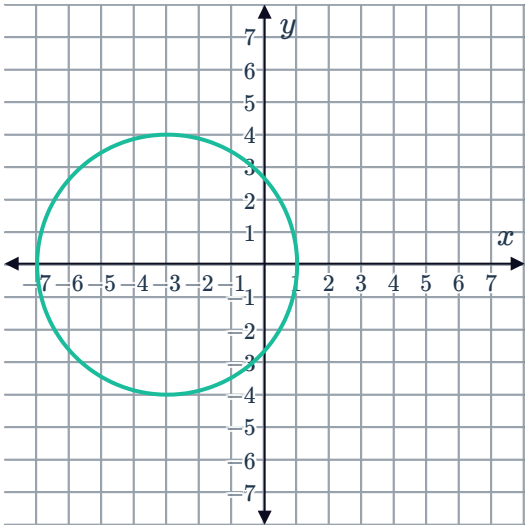
i $(-3, -1)$ ii 5 units

7

a

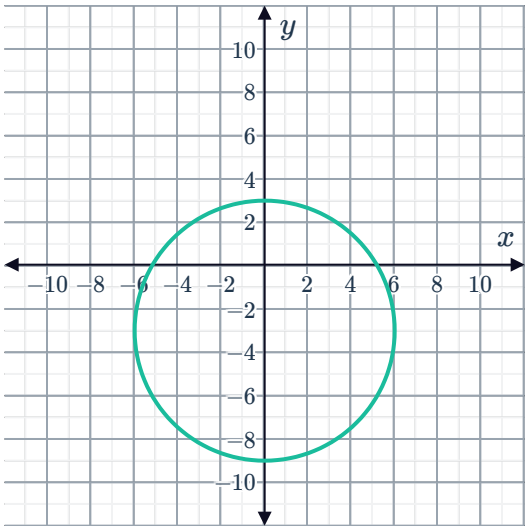


b



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c



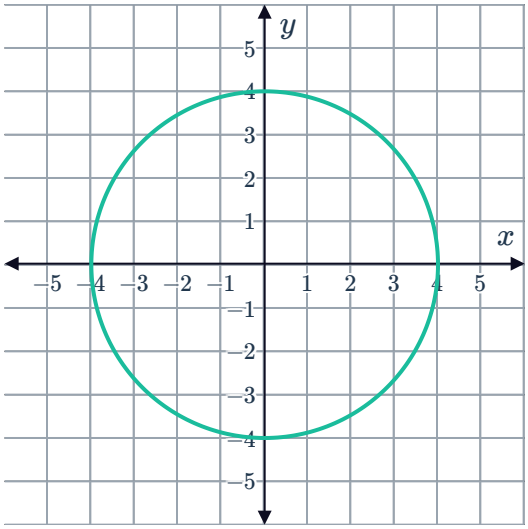
8

a

i $(0, 0)$

ii 4 units

iii

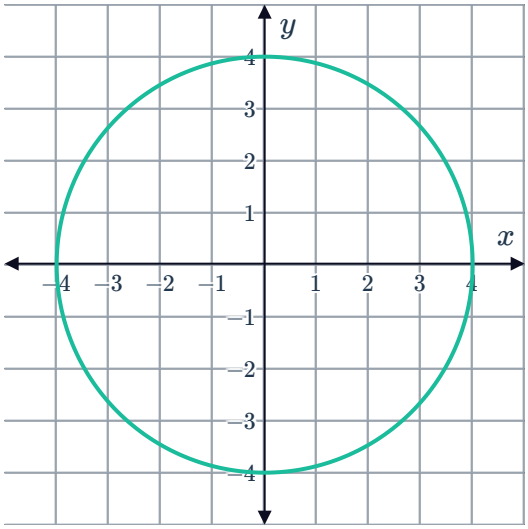


b

i $(0, 0)$

ii 4 units

iii



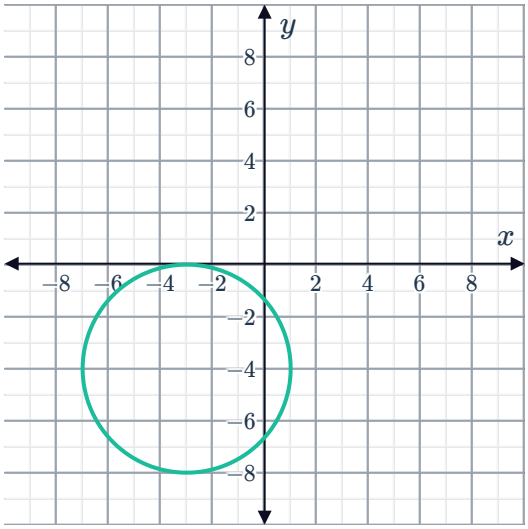
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c

i $(-3, -4)$

ii 4 units

iii

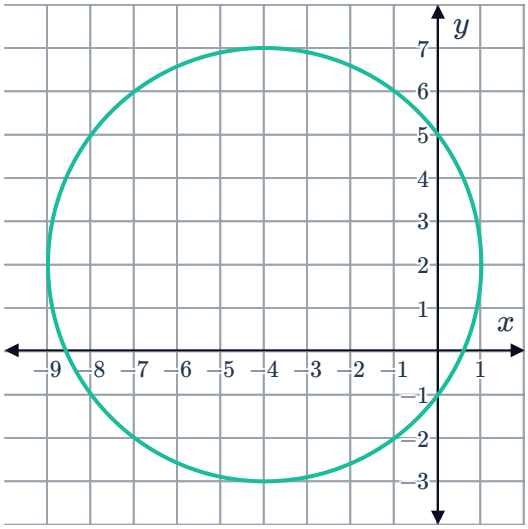


d

i $(-4, 2)$

ii 5 units

iii

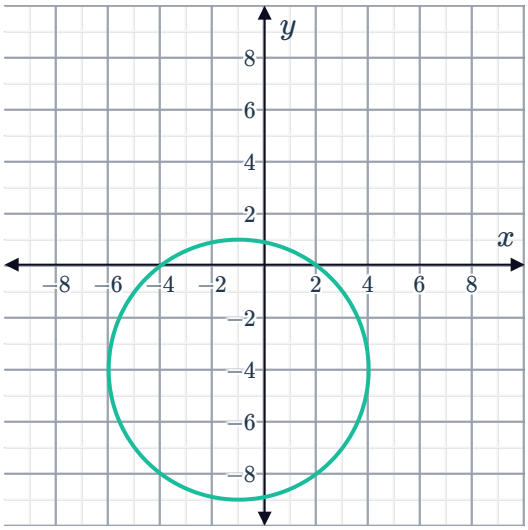


e

i $(-1, -4)$

ii 5 units

iii

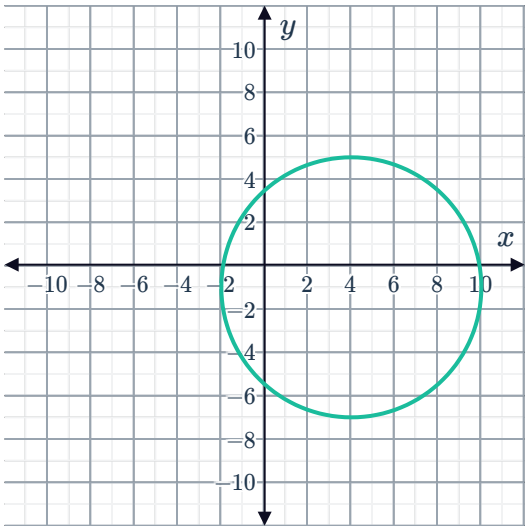


f

i $(4, -1)$

ii 6 units

iii



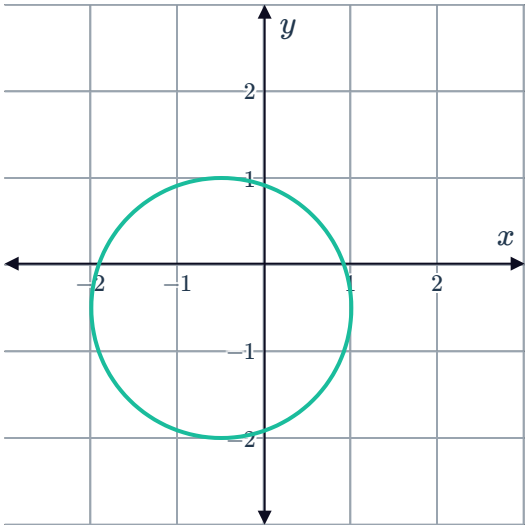
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g

i $\left(-\frac{1}{2}, -\frac{1}{2}\right)$

ii $\frac{3}{2}$ units

iii



9

a 41 units **b** No

10 $(7, -6)$

11

a 5 units **b** 5 units **c** $(5, -1)$ **d** Yes

12

a $(6, 5)$ **b** 2 units

Equations of circles

13

a No **b** No **c** No **d** Yes

e Yes **f** Yes

14

a $5\sqrt{5}$ units **b** $x^2 + y^2 = 125$

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15

a $x^2 + y^2 = 36$

b $36\pi \text{ units}^2$

16 **B**

17

a $(1, 3)$

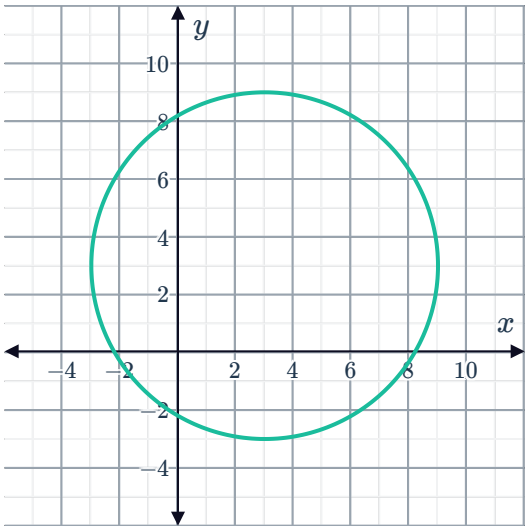
b 6 units

c $(x-1)^2 + (y-3)^2 = 36$

18

a

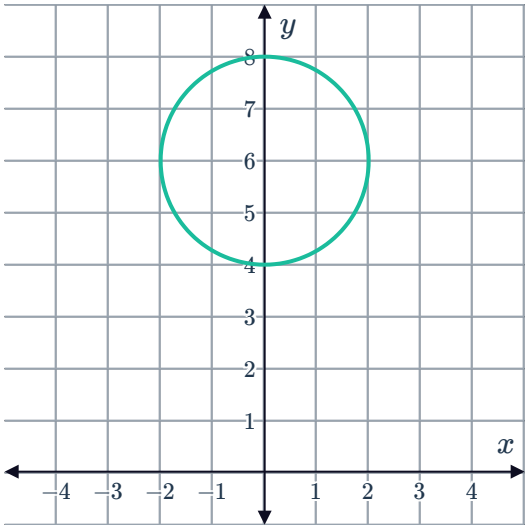
i



ii $(x-3)^2 + (y-3)^2 = 36$

b

i

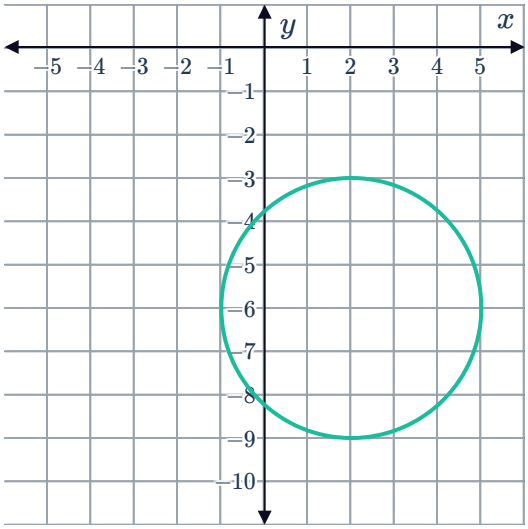


ii $x^2 + (y-6)^2 = 4$

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c

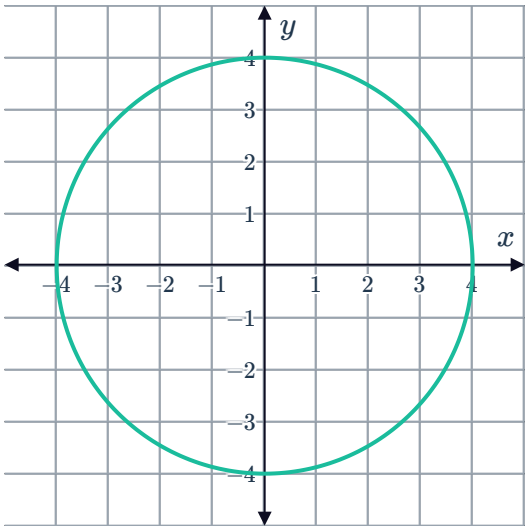
i



ii $x^2 - 4x + y^2 + 12y = -31$

d

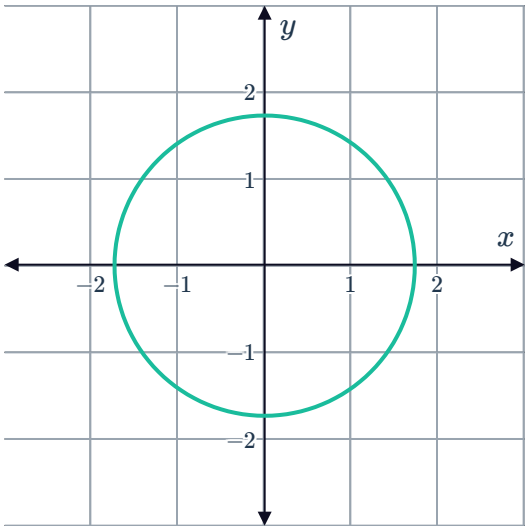
i



ii $x^2 + y^2 = 16$

e

i

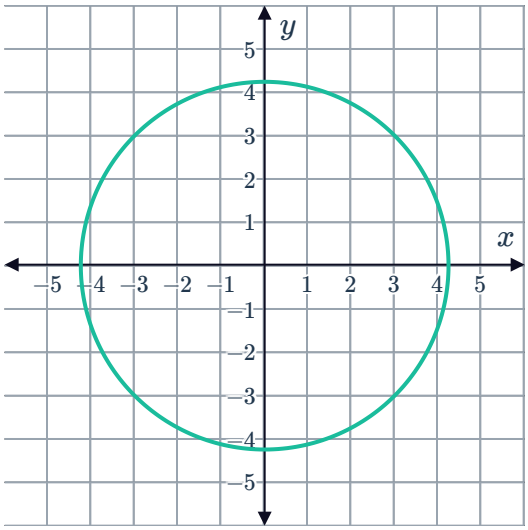


ii $x^2 + y^2 = 3$

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f

i



ii $x^2 + y^2 = 18$

19

a $\sqrt{74}$ units

b $(x+2)^2 + (y-3)^2 = 74$

20

a

i $(x-4)^2 + y^2 = 36$

ii $(4, 0)$

iii 6 units

b

i $(x+2)^2 + (y+3)^2 = 16$

ii $(-2, -3)$

iii 4 units

c

i $(x-1)^2 + (y+4)^2 = 36$

ii $(1, -4)$

iii 6 units

d

i $(x+4)^2 + (y-3)^2 = 1$

ii $(-4, 3)$

iii 1 units

21

a $\left(x - \frac{7}{2}\right)^2 + \left(y + \frac{5}{2}\right)^2 = 5$

b $(3.5, -2.5)$

c $\sqrt{5}$ units

d Inside

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22

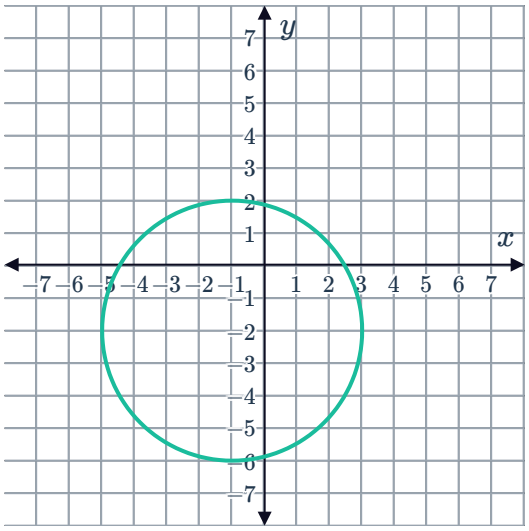
a

i $(x+1)^2+(y+2)^2=16$

ii $(-1,-2)$

iii 4 units

iv



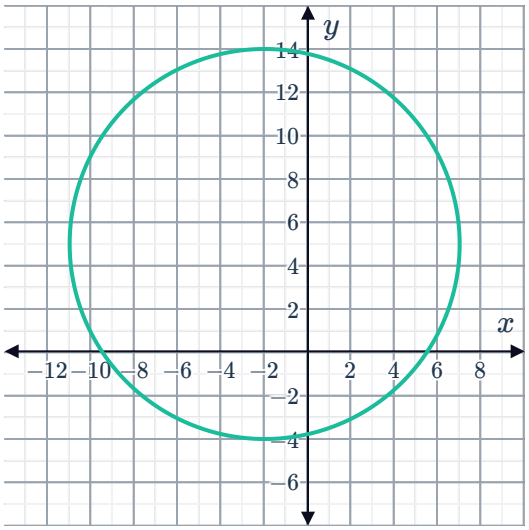
b

i $(x+2)^2+(y-5)^2=81$

ii $(-2,5)$

iii 9

iv



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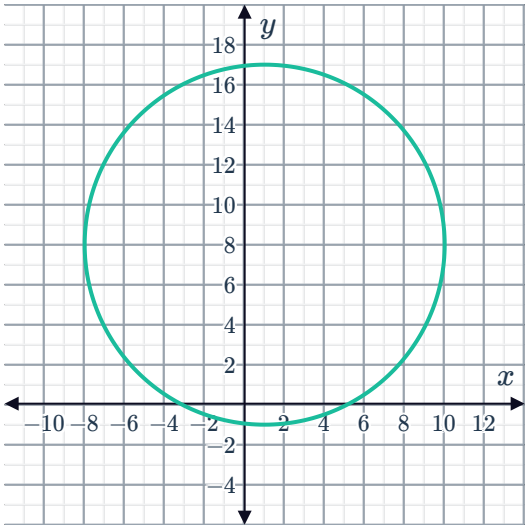
c

i $(x-1)^2+(y-8)^2=81$

ii $(1,8)$

iii 9

iv



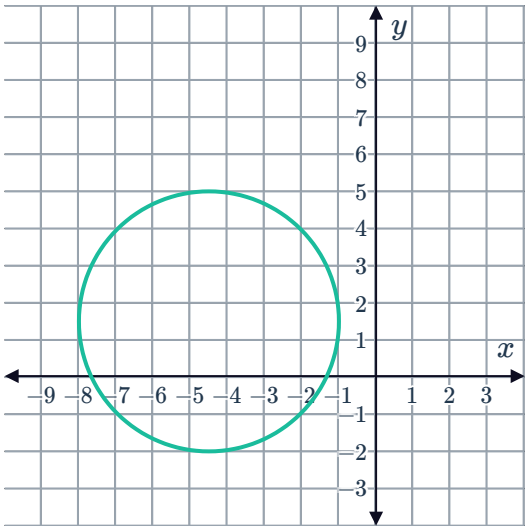
d

i $\left(x+\frac{9}{2}\right)^2+\left(y-\frac{3}{2}\right)^2=\frac{49}{4}$

ii $\left(-\frac{9}{2},\frac{3}{2}\right)$

iii $\frac{7}{2}$ units

iv



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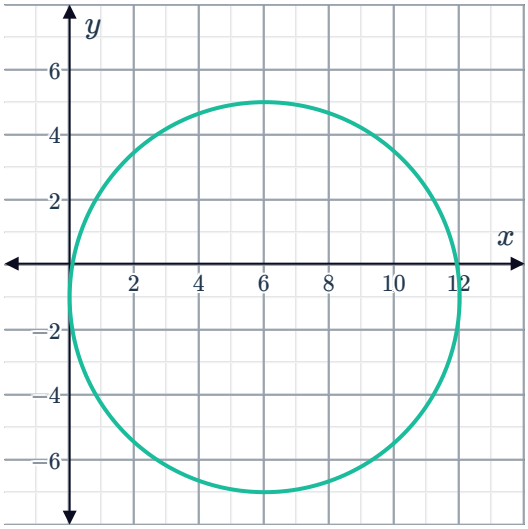
23

a $(x-6)^2+(y+1)^2=36$

b $(6,-1)$

c 6 units

d



24

a $(-10,-9)$

b $\sqrt{61}$ units

c $(x+10)^2+(y+9)^2=61$

Translations of circles

25 The graph of $(x+3)^2+y^2=4$ has been horizontally translated from the graph of $x^2+y^2=4$ by 3 units to the left.

26

a $x^2+(y+2)^2=25$

b $x^2+(y-3)^2=7^2$

c $(x-4)^2+y^2=9$

d $(x+5)^2+y^2=4$

Domain and range of circles

27

a

i $[-6,6]$ **ii** $[-6,6]$

b

i $[3,9]$ **ii** $[-4,2]$

28

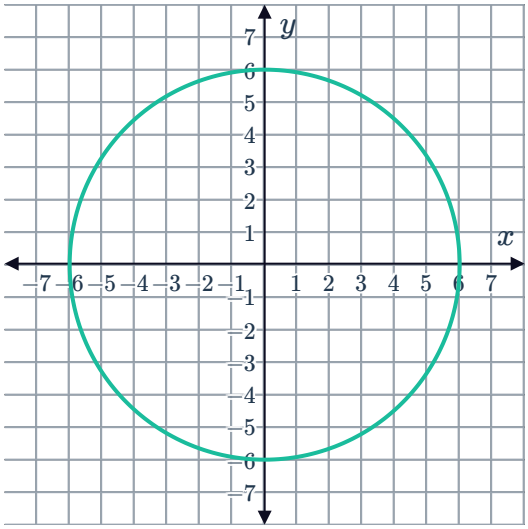
a $[-13.5,-10.5]$ **b** $[-14.5,-11.5]$ **c** Outside

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29

a

i

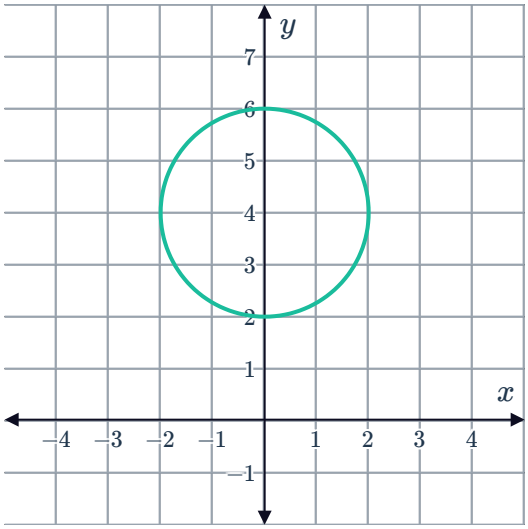


ii $[-6, 6]$

iii $[-6, 6]$

b

i



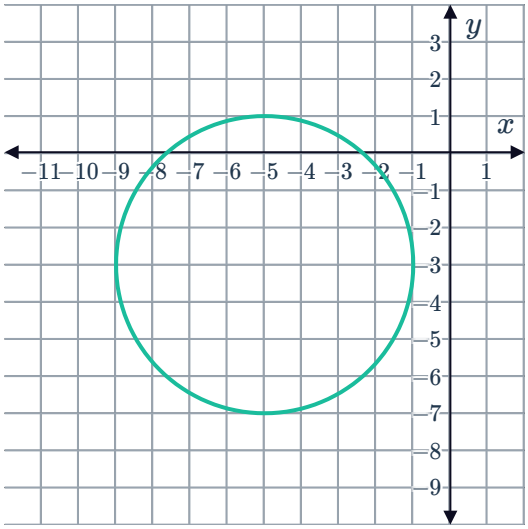
ii $[-2, 2]$

iii $[2, 6]$

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c

i

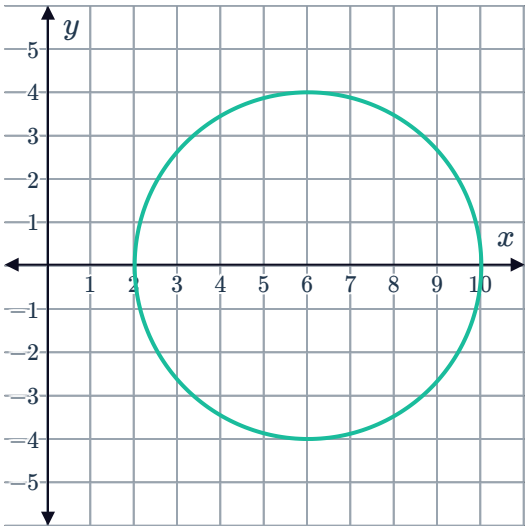


ii $[-9, -1]$

iii $[-7, 1]$

30

a



b $(x-6)^2 + y^2 = 16$

Applications

31

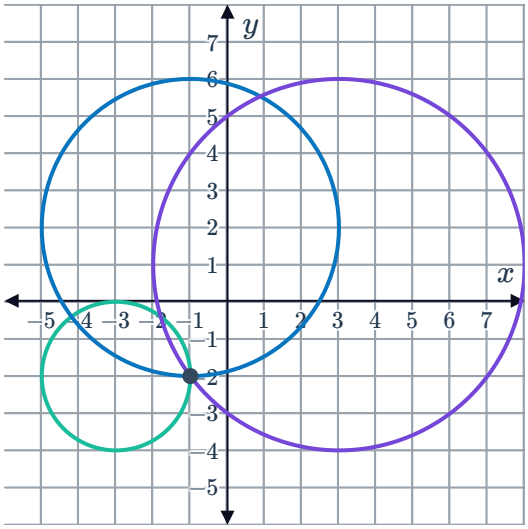
a $50\ \text{km}$

b No

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32

a As the point of intersection of the three circles centered at the receiving stations with radii as the distances, the epicenter is at $(-1, -2)$.



b The point $(8, -7)$ is outside the circle with center $(-1, -2)$ with radius 10, so it would not feel the earthquake.

33

a Domain: $[5, 19]$, Range: $[1, 15]$

b Domain: $[14, 26]$, Range: $[2, 14]$

c 5 cm

d 85.19 cm^2

e Circle one: $(x + 1)^2 + (y - 6.085)^2 = 49$, Circle two: $(x - 7)^2 + (y - 6.085)^2 = 36$

34 $x^2 + (y - 17)^2 = 144$

35 $(x + 5.3)^2 + (y + 5.4)^2 = 900$

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